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Airline ground operations: Optimal schedule recovery with uncertain arrival times

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ABSTRACT

Airline operations are subject to a number of stochastic influences which result into variable ground and block times for same flights on different days. Our research explores how airline operations control centers may benefit from an integrated decision support system for schedule recovery during aircraft ground operations. In this paper, we study the sensitivity of an optimal set of schedule recovery options towards uncertain arrival times. The calculation of recovery options is based upon an integrated and iterative scheduling and optimization algorithm, which incorporates uncertainties for arrival flights as a function of a given look-ahead time. Potential recovery options include stand re-allocation, quick-turnaround, quick passenger transfer, waiting for transfer passengers, cancellation of passenger or crew connections, and arrival prioritization. Within a case study setting at Frankfurt Airport, 20 aircraft turnarounds are analysed during a morning peak with their respective estimated arrival times (including potential arrival delays). The analysis of simulation results reveals an almost identical set of selected recovery options under high uncertainty circumstances and from post-operational point-of-view, which indicates high solution stability.

1. Introduction

Hub airports are central linking elements within the transport network of full-service network carriers (FSNC), given that many Origin-and-Destination (O&D) airline products involve multiple flights and connect via such an airport. Building a system around its actual users and their product structure (e.g., network operations) is key to achieve higher system performance and resilience in the event of schedule deviations. Resilience is defined as the capacity of a system not only to absorb, but to adapt to or restore a loss in system performance ((Hollnagel, 2013; Francis and Bekera, 2014; Cook et al., 2016)). Thus, the SESAR ATM Master Plan (Eurocontrol, 2019) strongly recommends the full integration of airports into the ATM network to facilitate airline operations. Understanding airline network operations involves a paradigm shift from gate-to-gate (i.e., flight-centric) towards air-to-air (i.e., hub-centric) perspective, so that interdependencies between arrival and departure flights at an airport can be accurately incorporated (Pilon et al., 2016).

1.1. Status Quo

Our research approach supports the SESAR vision for a holistic framework on ATM resilience and puts special focus on tactical airline schedule recovery. Given that airline operations are subject to various stochastic influences, tactical schedule recovery can be defined as the on-going monitoring and adjustment process of an airline's reference business trajectories (RBTs) on the day of operations, such that adaptive and restorative resilience capacities are utilized. In order to minimize the economic impact from deviations, airlines have established operation control centers (AOCC) for their schedule recovery, given that absorptive resilience capacities (implemented as strategic schedule buffers) do not suffice to mitigate all occurring deviation throughout an entire season. Current procedures within the AOCC are mainly based on expert-knowledge of the operating agents and require long negotiation times to define feasible solutions which adhere to pre-set recovery policies and incorporate the interests of all involved airline departments and local partners, such as air traffic control, airport and ground handling service providers (Ball et al., 2007).

In order to improve decision-making between agents, recent publications propose models which integrate recovery options for different

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flights and trajectory segments (Delgado et al., 2016; Marla et al., 2017; Mazzarisi et al., 2019). Thus, trade-offs are made between additional en-route fuel consumption to reduce arrival delays and delaying the departure of connecting flights to "wait-for-passengers". The proposed model in this paper intends to enhance the decision-making process within an AOCC by integrating rules and constraints from several departments into a common decision support system. Thereby, we focus on schedule recovery options for ground operations which are taken by the local airport unit of the AOCC, called Hub Control Center (HCC). The incorporation of potential recovery options which may be applied during airborne segments is left for future research. For the current model, we adopt elements from a wide range of research areas which so far have tackled the optimization of airline and airport operations with the isolated view of the respective stakeholder or department. These research areas comprise:

- Tactical Flight Prioritization;
- Prediction & Control of Turnaround Target Times;
- Aircraft-Specific Delay Metrics;
- Passenger-Oriented Delay Metrics;
- Tactical Crew Control Problem;
- Tactical Stand Allocation; and
- Assignment of Ground Servicing Equipment & Staff.

1.2. Focus and structure

The purpose of this study is to assess the sensitivity of decision variables (i.e., selected recovery options) towards uncertainties in the prediction of actual arrival times. During the flight, arrival times are frequently updated as Estimated In-Block Times (EIBT) and shared with all related stakeholders within the A-CDM database. Airlines typically have their individual prediction tools on top of A-CDM, however, this does not eliminate prediction errors related to weather, traffic and sensory conditions (Tielrooij et al., 2015). Thus, the sensitivity analysis is necessary to reveal whether secure information upon the optimality of schedule recovery decisions can be provided to the operating agents in the AOCC (the system users), given their limited situational awareness because of fluctuating EIBTs. Robust decisions are a necessary precondition to generate trust into a system and avoid further disruption arising from short-term schedule changes. Judging from the results, we evaluate whether future work on our model is needed to achieve robust decision-making (RDM) despite the frequent update of EIBT predictions. Sec. 2 of this article describes the integrated schedule recovery model in further detail, providing also references to related research areas listed above. Sec. 3 presents the implementation and application of the model to a case study environment with 20 turnarounds at Frankfurt Airport. In Sec. 4, the sensitivity of decision variables is analysed. Finally, Sec. 5 discusses the results, draws conclusions and provides an outlook onto future research steps.

2. Schedule recovery during aircraft ground times

As mentioned before and to the best of our knowledge, there has been no research about integrating all schedule recovery options available to airlines during aircraft ground times. The entire AOCC decision-making process, including all major departments, has been modelled in (Kohl et al., 2007; Clausen et al., 2010), however, aircraft-, crew- and passenger-related recovery options were distributed into separate modules, given that an integrated approach was not achievable due to the complexity of constraints. Another study complemented an aircraft recovery model with simplified crew, passenger and maintenance constraints (Bratu and Barnhart, 2006), leading the modelling path for integrated schedule recovery. Building on this (Petersen et al., 2012), aim at integrating the solution from several models, all concurrently minimizing the cost for schedule recovery as well as aircraft, crew and passenger recovery. They concluded that (especially crew) constraints need to be simplified and passenger itinerary alternatives need to be pre-selected to achieve a globally optimal solution within a close to real-time solution environment. Furthermore, if the majority of flights was impacted by a network disruption, problem complexity exceeded the capacity to find an integrated optimal solution of all models. Other publications, which focus on airline schedule recovery, widely consider ground operations (i.e., the turnaround) as a singular process which has a stochastic duration (Stojković et al., 2002; Rosenberger et al., 2002; Wu and Law, 2019). While the stochastic approach has been proven superior in comparison to initial studies, which assumed static ground times, it still neglects many resource dependencies at the airport which can cause delay propagation across multiple aircraft. According to the Eurocontrol performance review report 2019, ground operations have been the number one source of primary delay to the ATM system, causing about 33% of flight departure delays (Eurocontrol, 2020). Thus, integrated airline schedule recovery models should incorporate further constraints considering the decision-making in the local hub control center.

Most studies which actually acknowledge the complexity of ground operations, model the turnaround as a network of interdependent sub-processes. Thereby, the majority focuses on one individual turnaround while building stochastic simulation models for target time predictions (Wu and Caves, 2004; Schlegel, 2010; Oreschko et al., 2012; Schultz et al., 2013) or proposing resource-constrained scheduling models with incorporated process alternatives (Kuster et al., 2009). However, also these studies neglect potential resource dependencies between multiple turnarounds at the same airport. Such interdependencies are modelled for individual resources, such as for aircraft parking positions and ground handling equipment (Vidosavljevic and Tosic, 2010; Padrón et al., 2016; Dorndorf et al., 2017; Dijk et al., 2019; Tomasella et al., 2019), pushback and towing trucks (Du et al., 2014), de-icing positions (Norin et al., 2012) and taxi-ways (Mota et al., 2017). These models aim for an optimal resource allocation from the operators perspective, largely omitting the airline network perspective with its complex crew duty constraints and passenger itineraries, which result in increasing reactionary cost with higher departure delay (Beatty et al., 1999; AhmadBeygi et al., 2008; Campanelli et al., 2016). Our approach, therefore, seeks to mediate between the macroscopic airline network and microscopic ground operations perspectives, such that it brings elements from both worlds together into an integrated schedule recovery model for airline ground operations.

2.1. Integrated schedule recovery model for parallel turnarounds

The integrated turnaround monitoring and scheduling tool, "Ground Manager - GMAN 2.0" (Evler et al., 2018, Evler et al., 2021), models multiple parallel aircraft turnarounds and is applied with a rolling planning horizon within this article. Each aircraft turnaround $a \in A$ is defined by a scheduled start and finishing time of the turnaround $SIBT_a$ and $SOBT_a$ which are retrieved from the flight plan. For a pre-defined number of iterations within the rolling horizon, estimated arrival times $EIBT_a$ are predicted for each aircraft a based on historical flight data and are used to calculate target off-block times $tobt_a$ and recovery decisions as the result of the scheduling process. The turnaround of all aircraft includes a network of sub-processes TP, where each sub-process $i \in TP$ has a variable starting time s_i, a duration d_i, is characterized by the related aircraft RA_i and by links to preceding and succeeding processes determined in the precedence matrix $PM \subseteq TP \times TP$ (see Fig. 1).

Earlier studies (Fricke and Schultz, 2009; Oreschko et al., 2012; Schultz et al., 2013) have analysed stochastic process durations and found case-sensitive probability density functions for each turnaround sub-process. From these fitted distributions, we adopt mean values as deterministic parameters with regard to the respective aircraft type, airport and flight characteristics. For instance, the cleaning duration d_{CL} for an Airbus A320 aircraft is derived as the mean value from a set of functions (1) determined by the trigger parameters aircraft-type A/C_{type},

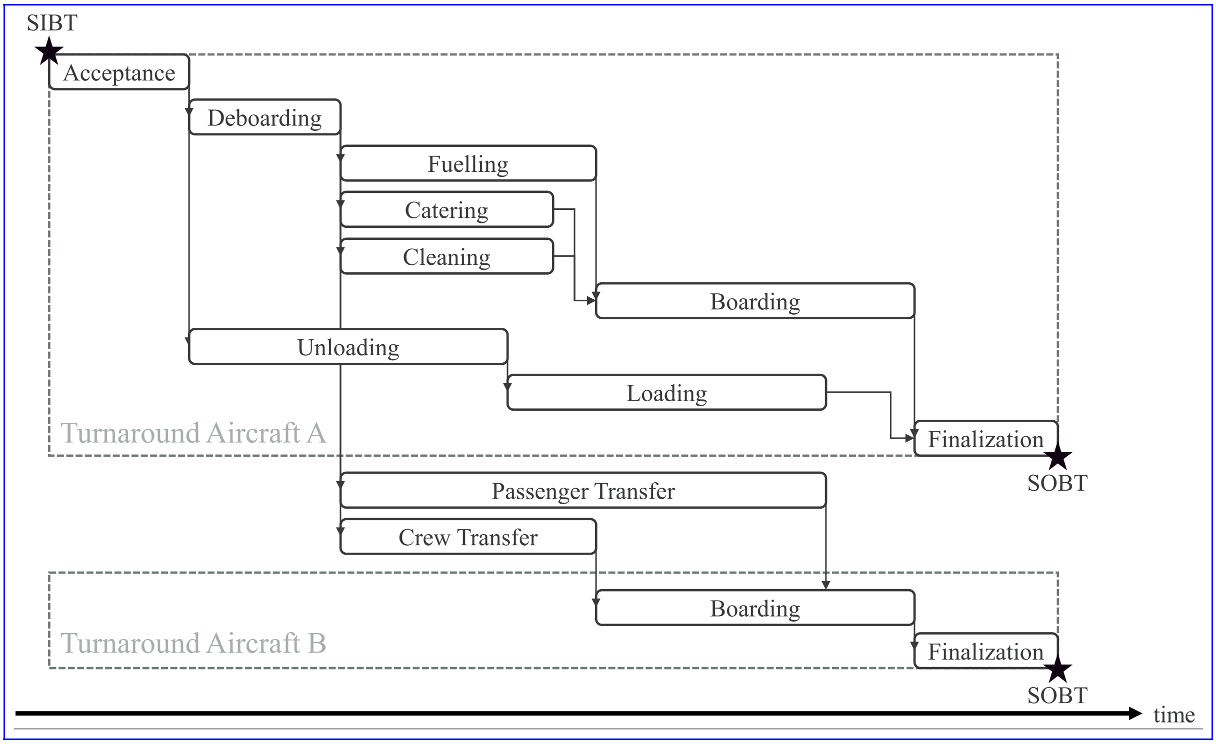


Fig. 1. Network graph of turnaround sub-processes.

seat-load-factor SLF_{ARR} and distance $Dist_{ARR}$ of the arrival flight (ARR - cf. Fig. 2). Thus, contrarily to the description in ground operations manuals, the critical path is flexible and can contain different processes for different turnarounds, so that the estimated total turnaround time might deviate from official minimum ground times even in case of on-time arrivals (Evler et al., 2018).

$$d_{CL} = f(A / C_{type}, SLF_{ARR}, Dist_{ARR}) \tag{1}$$

The model itself is an extension of the Resource-Constraint Project Scheduling Problem (RCPSP) which considers turnaround processes as jobs and airport resources as machines (RCPSP is the general framework of shop scheduling models (Nasiri, 2013)). While each aircraft needs to be assigned to an airport stand, which is associated with the necessary equipment and personnel for a standard turnaround (typcial RCPSP), sub-process sequences can be altered and some processes can be accelerated or eliminated depending on the availability of additional airport resources (extended RCPSP - Kuster et al., 2009; Deblaere et al., 2011). Furthermore, the extended model permits milestones like the scheduled off-block time $SOBT_a$ to be overrun. The resulting delay v_a for aircraft a is measured with increasing marginal cost of delay C_a^{DD}, which have been determined for the respective aircraft types in (Cook, 2015). Considering arrival delay ($EIBT_a > SIBT_a$) as a deviation to the original flight plan, the model introduces standardized schedule recovery options RO which allow to change the operating procedure for certain sub-processes and,

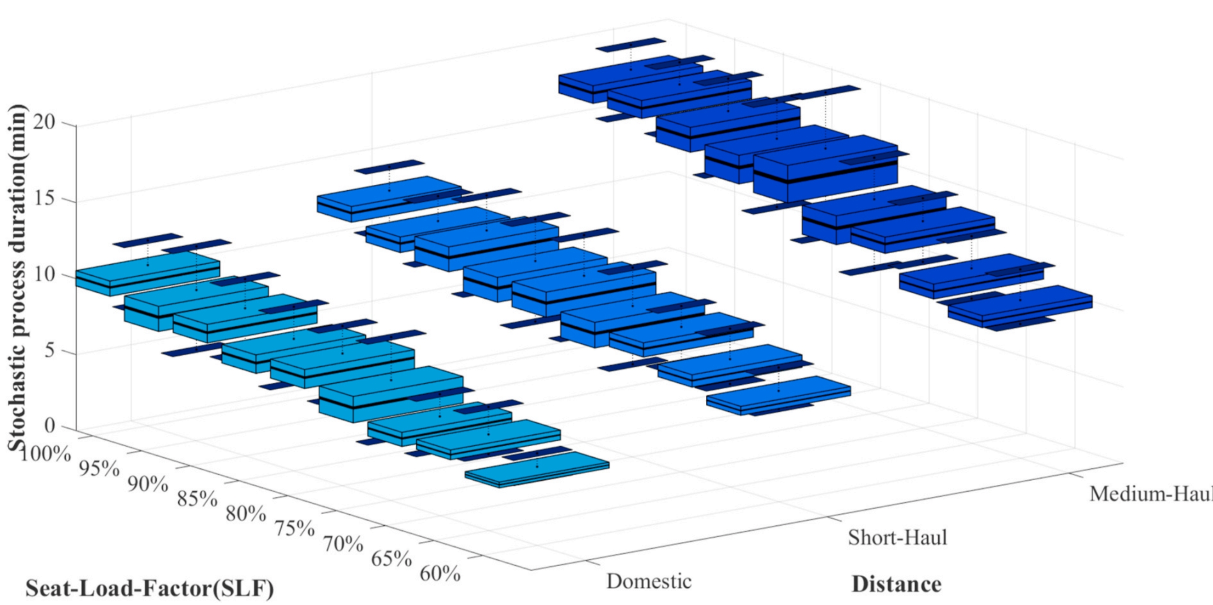
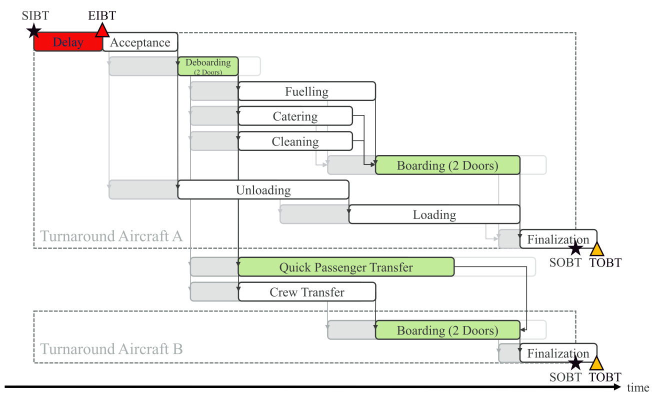


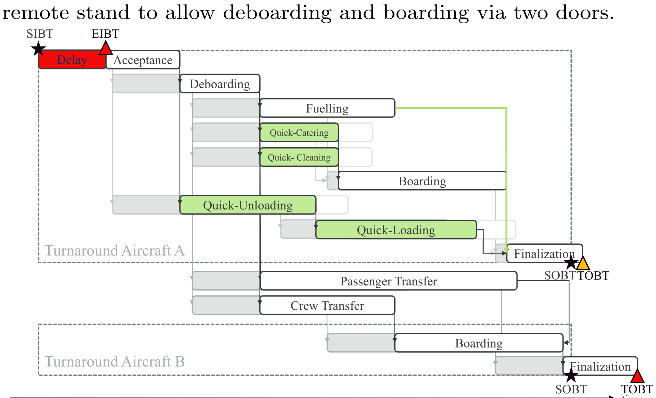
Fig. 2. Multivariate duration of A320 cleaning process.

thus, influence the $tobt_a$.

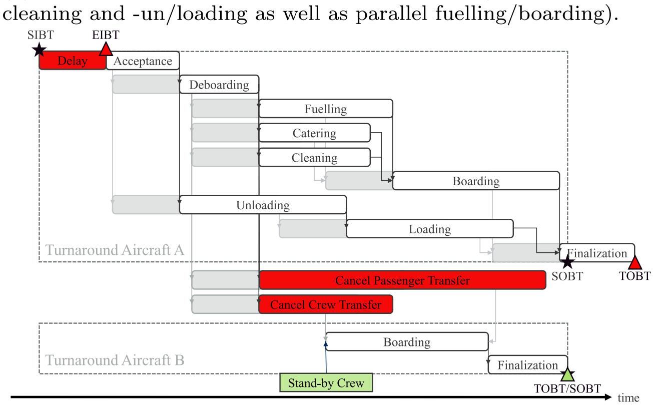
Recovery options can be categorized into inbound recovery, transit recovery and outbound recovery. Inbound recovery is applied in order to guarantee passenger or crew transfer connections which might be disrupted or cause reactionary departure delays for other aircraft due to an arrival delay (see Fig. 3a). Thus, the initial stand allocation can be adapted such that a delayed aircraft can be re-positioned from a contact stand at the terminal to a remote stand on apron, which allows quicker deboarding (and boarding) procedures as more doors can be used simultaneously (Schultz, 2018, 2018a). Additionally, quick-transfer busses can be assigned to accelerate the needed transfer time NTT_{kl} for passengers (see Fig. 3a). Another considered inbound recovery option is arrival prioritization, where the airline can request ATC to give priority to the approach or taxi-in process of an aircraft with the benefit to cut some minutes off the initial arrival delay.



(a) Inbound recovery with quick transfer and re-allocation to an remote stand to allow deboarding and boarding via two doors.



(b) Transit recovery with quick-turnaround (quick-catering, -cleaning and -un/loading as well as parallel fuelling/boarding).



(c) Outbound recovery with cancelled passenger and crew transfer connections.

Fig. 3. Overview of potential schedule recovery options in case of arrival delay.

Transit recovery is supposed to restore the aircraft rotation of the delayed arrival flight. A typical recovery option includes the prioritization of aircraft servicing processes for aircraft a ($\omega_a = 1, 0$ otherwise), so that additional personnel complements the already assigned cleaning, catering or loading crews on delayed turnarounds in order to reduce the duration of the respective sub-processes by a factor α. Furthermore, the airline can request whether fuelling can be operated during the passenger boarding process (normally both processes run sequentially), which requires authorization by the local fire brigade and the aircraft captain. The highest impact is achieved when all mentioned sub-processes of one turnaround are prioritized simultaneously – which is called quick-turnaround. This was found to guarantee a reduction of the $tobt_a$ even under stochastic influences (Evler et al., 2018) (see Fig. 3b).

Outbound recovery goes hand in hand with inbound recovery, as aircraft which have no initial arrival delay can be held on position to avoid misconnections (see Fig. 3b), while the airline might also decide to cancel connections ($\varphi = 1$, 0 otherwise) by substituting the planned crew with a standby reserve or by rebooking passengers PA onto other flights (see Fig. 3c). This would eliminate non-rotational reactionary delay which might spread to multiple stations in the airline network. Related cost for passenger care, rebooking, lodging and compensation (C_i^{CP}), which depend on the trip length and delay at final destination, are adopted from extensive research at European level (Cook and Tanner, 2015).

Note that all schedule recovery options can be freely combined with each other for one turnaround as long as the respective resources are available to perform them without any further disturbances. Thereby, ground service equipment (GSE), staff and airport stands require certain transition times TT_{ab} between two subsequent assignments (see Fig. 4), which requires sequencing constraints (Padrón et al., 2016; Tomasella et al., 2019). The number of available (extra) units or assignments per operational period N^{RO} is usually defined within service-level-agreements between airline and airport or ground handler. As mentioned above, our scenario assumes each turnaround to be associated with a set of standard resources when it is assigned to an aircraft stand, so that on top of that, the model only routes extra resources to accelerate standard procedures.

Furthermore, the allocation of aircraft a to stand k ($\chi_{ak} = 1$, 0 otherwise) is constrained by aircraft type (i.e., size) and security regulations (i.e., Schengen or Non-Schengen) which depend on the countries of arrival and departure flights. One also needs to consider that needed transfer times NTT_{kl} from stand k to stand l change for all connecting passengers to their respective on-wards flights when an arrival flight is re-allocated to another stand and vice versa for passengers which transfer to the re-allocated aircraft. Such complexities are typically summarized in isolation within the so-called Tactical Stand Assignment Problem (TSAP), which can differentiate between airline or airport objectives (Dorndorf et al., 2017; Dijk et al., 2019).

2.2. Mathematical formulation

Sets:	
A	set of all aircraft
TP	set of all turnaround sub-processes
IB	sub-set of *TP*, containing all aircraft acceptance processes
CL	sub-set of *TP*, containing all aircraft cleaning processes
FI	sub-set of *TP*, containing all aircraft finalization processes
PA	set of all passenger transfer processes
ST	set of all aircraft stands
AP	start and end node for sequencing of recovery resources
Parameters:	
C_a^{DD}	cost of departure delay of aircraft *a*
C_a^{RO}	cost of recovery option for aircraft *a*
C_i^{CP}	cost of cancelled transfer connection for passenger group *i*
N^{RO}	number of available resources for recovery options
d_i	duration of sub-process *i*
NTT	needed transfer time between stands *k* and *l*
α	time reduction factor achieved by recovery option
TT_{ab}	transition time between aircraft *a* and *b*
SIBTa	given scheduled in-block time for aircraft *a*
EIBTa	updated estimated in-block time for aircraft *a*
SOBTa	given scheduled off-block time for aircraft *a*
RA_i	related aircraft to turnaround sub-process *i*
M	context-specific big M
Variables:	
s_i	starting time of process *i*
v_a	departure delay for aircraft *a*
tobta	target off-block time for aircraft *a* based on the scheduling process
ω_a	binary variable – equal to 1 if recovery option applied to aircraft *a*, and 0 otherwise
φ_i	binary variable – equal to 1 if passenger transfer connection *i* is cancelled, and 0 otherwise
xQab	binary flow variable – equal to 1 if recovery option for aircraft *b* directly succeeds recovery option of aircraft *a*, and 0 otherwise
χ_{ak}	binary variable – equal to 1 if aircraft *a* is assigned to stand *k*, and 0 otherwise

$$\min \quad \sum_{a\in A}\left(C_a^{DD}\, v_a \;+\; C_a^{RO}\,\omega_a\right) + \sum_{i\in PA} C_i^{CP}\,\varphi_i \tag{2}$$

$$\text{s.t.} \quad s_i \geq EIBT_a \forall\, i \in IB | RA_i = a \tag{3}$$

$$s_i + d_i \leq SOBT_a + v_a \forall\, i \in FI | RA_i = a \tag{4}$$

$$s_j \geq s_i + d_i \begin{array}{l} \forall\, i \in \{TP \setminus CL, PA\}; \\ \forall\, j \in TP | (i,j) \in PM \end{array} \tag{5}$$

$$s_j \geq s_i + NTT_{kl}\,\chi_{ak}\,\chi_{bl} - M\varphi_i \begin{array}{l} \forall\, i \in PA;\ \forall\, j \in TP|(i,j) \in PM, \\ RA_i = a,\ RA_j = b;\ \forall\, k,l \in ST \end{array} \tag{6}$$

$$s_j \geq s_i + d_i(1-\omega_a) + \alpha\, d_i\, \omega_a \begin{array}{c} \forall\, i \in CL; \forall\, j \in TP; |(i,j) \in PM, \\ RA_i = a \end{array} \tag{7}$$

$$\sum_{a\in\{A,AP\}} xQ_{ab} = \omega_b \forall\, b \in A \tag{8}$$

$$\sum_{b\in\{A,AP\}} xQ_{ab} = \omega_a \forall\, a \in A \tag{9}$$

$$s_i \geq tobt_a + TT_{ab} - M(1 - xQ_{ab}) \begin{array}{c} \forall\, a \in A;\ \forall\, b \in \{A, AP\}; \\ \forall\, i \in AC | RA_i = a \end{array} \tag{10}$$

$$\sum_{b\in\{A,AP\}} xQ_{ab} \leq N^{RO} \forall\, a \in \{A, AP\} \tag{11}$$

The objective function (2) is to minimize schedule recovery cost related to a deviation from schedule, which corresponds to the sum of cost for departure delay and applied recovery options across all aircraft as well as cost for rebuilding passenger and crew connections. Thereby, the start of each turnaround can only be scheduled after the respective estimated in-block time (3). If the calculated target off-block time overruns the SOBT, the delay is induced as described in (4). Scheduling constraints (5) ensure that each sub-process can only start once all preceding processes have been finished. Constraints (6) schedule passenger transfer processes, depending on the stand allocation of the arrival and the departure flight and whether the connection is cancelled. Constraints (7) show exemplary for the cleaning process, how sub-processes can be accelerated by the application of recovery options. Sequencing constraints (8-10) ensure that recovery resources are assigned to only one turnaround at a time, while constraints (11) limit the number of available recovery resources.

3. Case study implementation with uncertain arrival times

Our case study consists of 20 turnarounds during a morning peak at Frankfurt Airport (FRA). The respective flight plan data are adopted from the summer schedule 2017 of a local hub carrier. Passenger connections are simulated in order to resemble potential itineraries which adhere to the average connection index (55% transfer passengers) and minimum connecting time in FRA (45 min), typical airline SLFs (85%) and avoid extreme detours (i.e., a passenger from Malta is unlikely to connect via FRA to Milan).

Crew connections and stand allocations have been simulated once such that they comply with official operational constraints. Thereby, contact stands in Terminal 1A (Stands A1–A5 in Fig. 5) are reserved for flights from and to Schengen countries only. Contact stands with special security and customs areas (Stands A6, B1-2, C1-2) can operate also flights from and to Non-Schengen countries, stands A6 and C2 being predominantly used for intercontinental flights with wide-body aircraft. Stands R1 to R3 are remote stands, where passengers need to be transferred with apron busses via the central bus station (marked with the bus icon in Fig. 5). Ten aircraft are fixed on their initially assigned stands

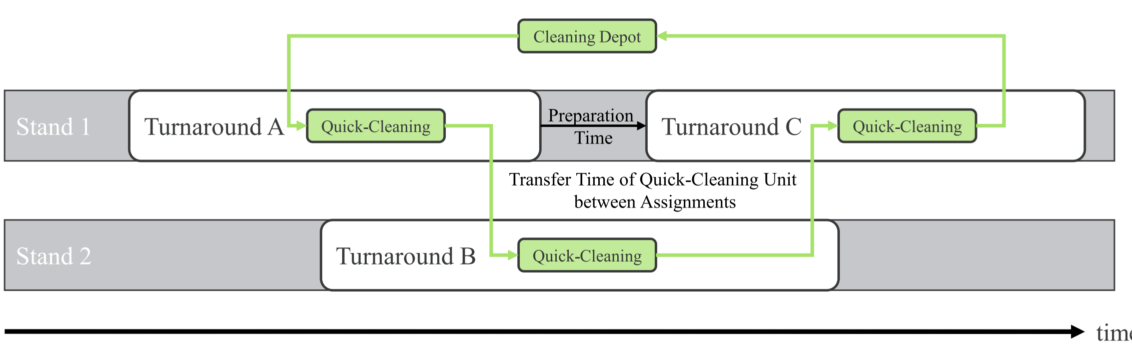


Fig. 4. Routing constraints of airport recovery resources.

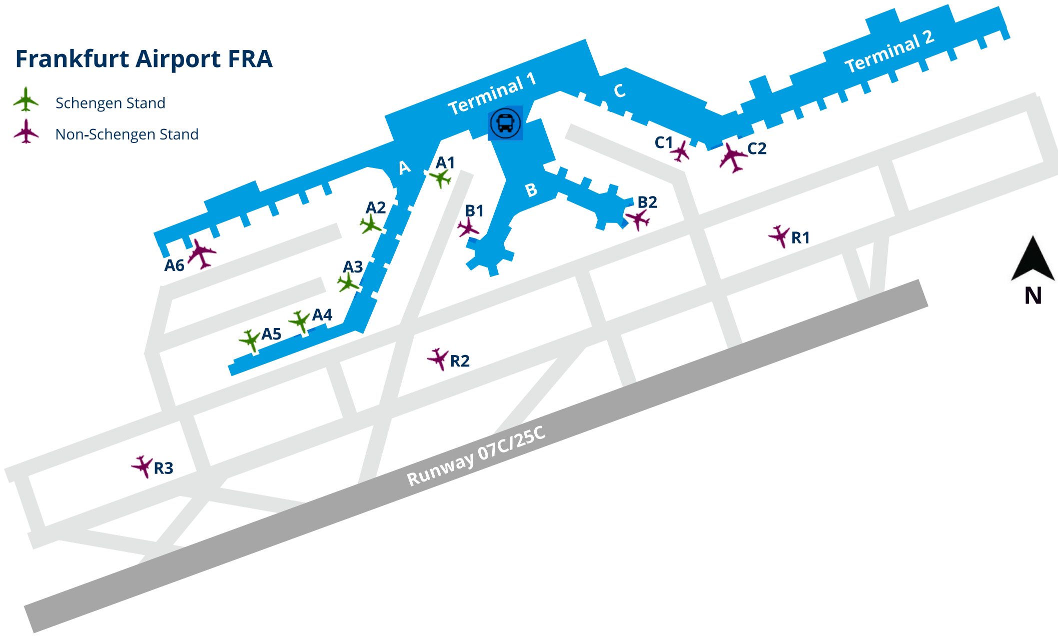


Fig. 5. Map of Frankfurt Airport with location of aircraft stands.

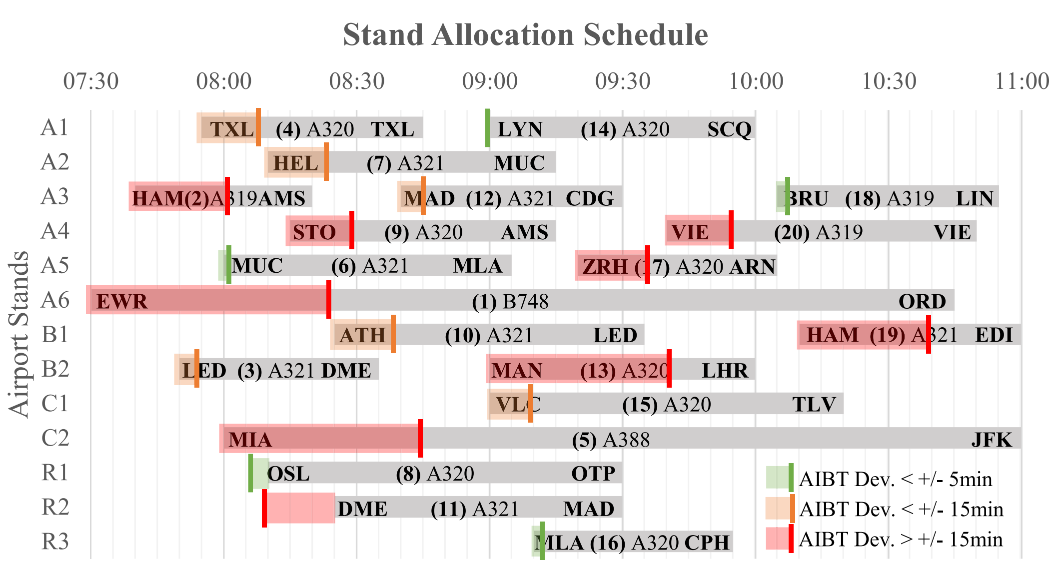


Fig. 6. Stand Allocation Scheme of aircraft in case studyGrey bars contains origin airport, aircraft number, aircraft type and destination airport, coloured bars exhibit deviation of AIBT from SIBT on the studied day. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

due to operational constraints, whereas the remaining ten aircraft can be re-allocated to any other stands which corresponds to the required security procedures of the origin and destination countries. In total, one arrival prioritization request, one quick-turnaround unit, one quick passenger transfer bus and one standby crew are available to enable the respective schedule recovery options.

Fig. 6 displays the initial stand allocation from the day of the case study with the respective actual in-block times (AIBT) of all inbound flights. The day contained thunderstorms North and West of FRA, so that some flights were delayed by up to 55 min. Two flights, however, arrived earlier than scheduled, resulting into an average arrival delay of about 15 min.

Assuming a tactical setting on the day of operations in the HCC, AIBTs (only known in retrospective) would be frequently predicted as EIBTs for the operating agents in the AOCC. Based on the constellation of EIBTs for all arriving aircraft within a certain period, AOCC agents would then aim at elaborating a cost-minimal solution for given schedule deviations with the help of the proposed decision support system "GMAN 2.0". However, research has found EIBT uncertainties to be increasing with longer look-ahead time (LAT) before actual arrival. Thus, when an aircraft is still on stand of its origin airport, EIBT uncertainties are equivalent to Weibull-distributed departure delays, while during the flight, they are normal distributed with decreasing variance towards arrival (Tielrooij et al., 2015). This means that AOCC agents, as well as the decision support system, base their calculations on a restricted situational awareness which is caused by EIBT uncertainties.

Hence, it is the purpose of this study to analyse the impact of this reduced situational awareness, if agents would always rely on the accuracy of the predicted EIBT values. Furthermore and given that all recovery options listed above involve resources, a certain freeze horizon needs to be considered in the decision-making process, so that an optimal solution can be implemented without causing further disturbances to the original plan.

Therefore, we extend the integrated schedule recovery model presented in Sec. 2 into an iterative optimization algorithm which incorporates uncertain EIBTs and is triggered according to pre-defined time-intervals T. The algorithm is repeated for 100 times ($N = 100$) to reach statistical significance. Time-intervals between iterations IT of the algorithm are determined to be one hour, which results into three iterations at 7:00 a.m.($t1$), 8:00 a.m.($t2$) and 9:00 a.m.($t3$). At the start of each iteration, AIBT values of all aircraft A are adopted as mean values from field data and assigned with different standard deviations depending on the look-ahead time LAT until arrival. Thereby, EIBT deviations are generated randomly according to a normal distribution around the mean values (AIBT) and have increasing variance ($\Sigma = 1.5; 2; 4$ minutes) with longer look ahead times. Within the rolling horizon, from iteration to iteration, EIBT uncertainty for each aircraft is then decreasing until arrival (see Fig. 7). Once EIBT values are updated, the schedule recovery model is run and the calculated optimal recovery options Ω_a are fixed for those aircraft that arrive within the next time-interval plus 15 min hold-over time (i.e., freeze horizon). Consequently, all flights arriving before 8:15 a.m. cannot be controlled any

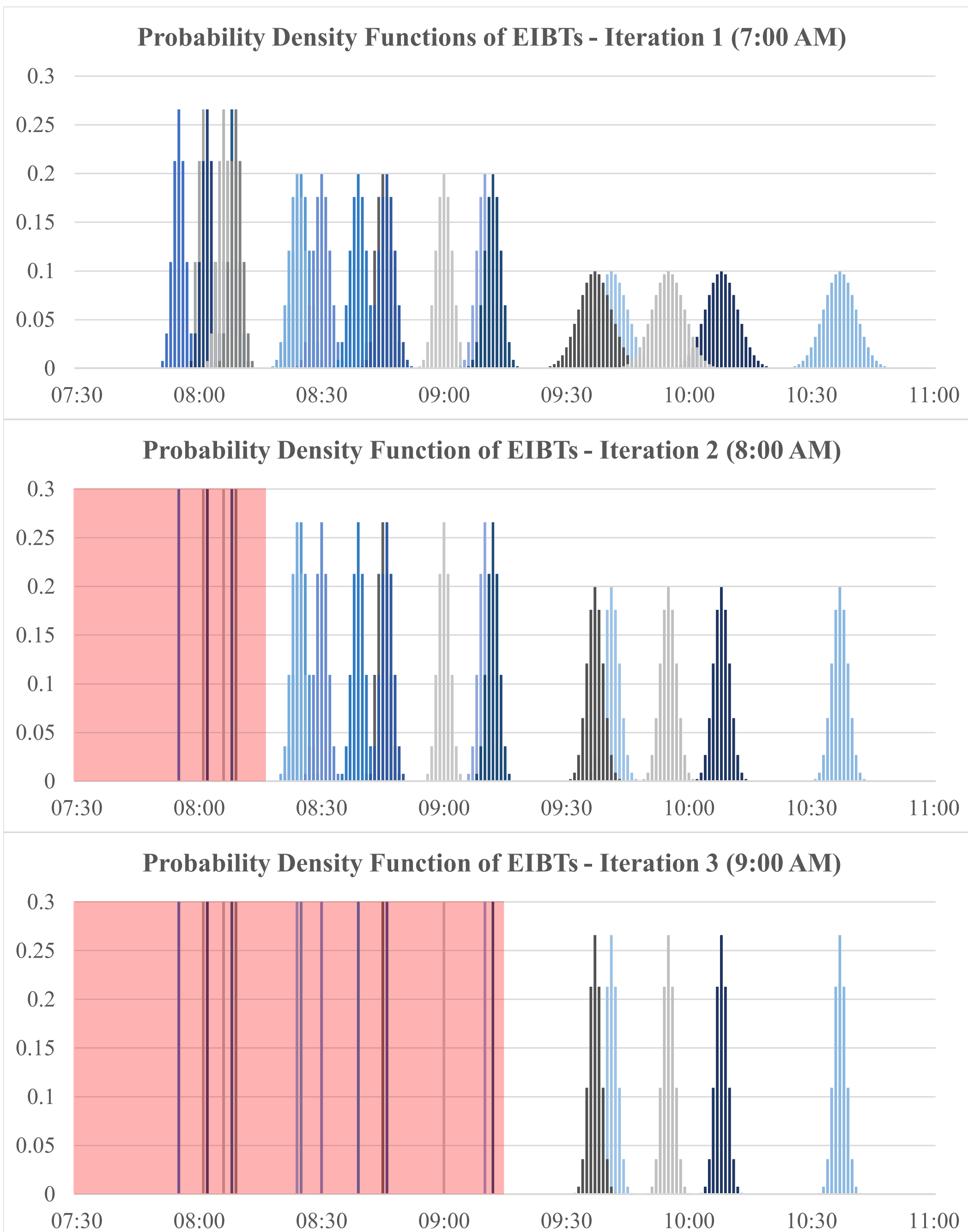


Fig. 7. EIBT uncertainties per iteration according to look-ahead times until arrival. Decision variables in the shaded areas in the second and third graph are fixed and cannot be changed any longer.

longer in t2 (at 8:00 a.m.) to allow enough implementation time for the recovery options. The same is valid for all decision variables of flights arriving before 9:15 a.m., which can no longer be controlled at t3 (see Algorithm 1). Finally, we compute the optimal solution with constant deterministic arrival times ($t4$ - AIBTs as exhibited in Fig. 6) and compare it to the results from the simulation.

Algorithm 1. Iterative Schedule Recovery.

Algorithm 1 Iterative Schedule Recovery

Input : AIBT, N, IT, LAT, A, T, Σ

Output: Updated EIBT, Optimal Set of Recovery Options Ω_a

```
for n ← 1 to N do
    for it ← 1 to IT do
        for a ← 1 to A do
            for l ← it to LAT do
                if AIBT_a ≤ t_l then
                    EIBT_a ← AIBT_a + σ_l
                end
            end
        end
        Run integrated schedule recovery model
        for a ← 1 to A do
            forall elements of Ω_a do
                if AIBT_a ≤ t_it + 15min then
                    Ω_a ← fix as parameter in the next iteration (it + 1)
                end
            end
        end
    end
end
```

that predicted EIBT values in each iteration are accepted as accurate.

As one might expect, schedule recovery costs have their largest variance in the first iteration which has the longest LAT and, thus, highest uncertainty. In fact, objective values return to be up to 23% higher but also 18% lower than optimal cost values determined with constant EIBT values (t4, marked as 0%). With decreasing EIBT uncertainty, the cost distributions asymptotically approach the optimum from t4, but are left-skewed towards higher average cost (see Fig. 8).

4. Case study analysis

As described in Sec. 3, the case study analyses the sensitivity of controller decisions with the presented model based on the limitation

4.1. Sensitivity of decision variables

To derive the sensitivity of decision-making with different levels of EIBT prediction accuracy, we compare the obtained decisions in each

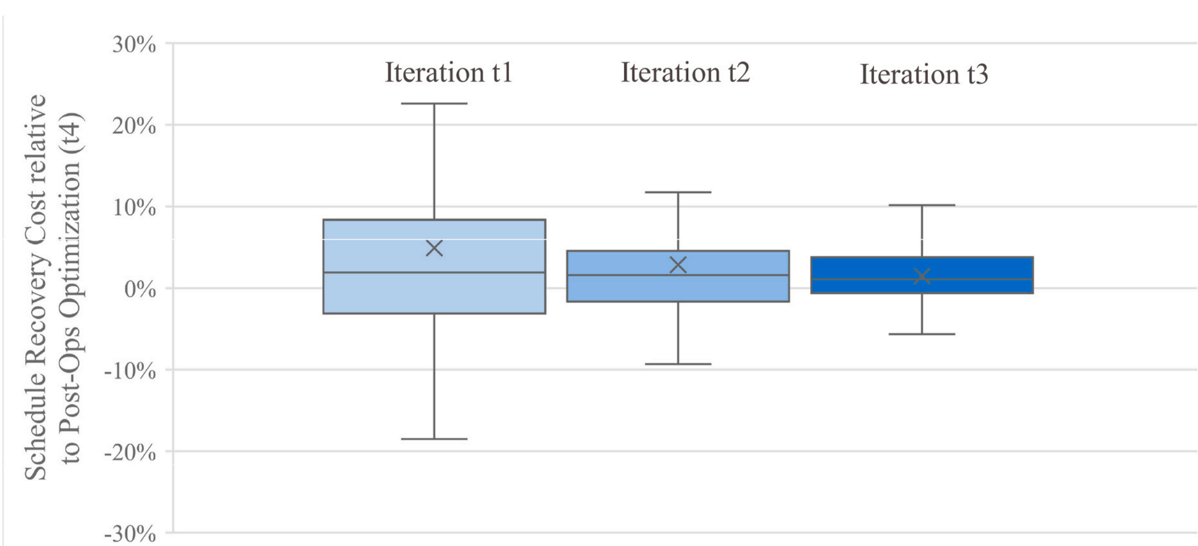


Fig. 8. Comparison of optimal schedule recovery cost in iterations $t1$, $t2$ and $t3$ (100 simulation runs with EIBT uncertainty) with optimal schedule recovery cost from post-operational perspective $t4$ with known AIBT (marked as 0%).

Table 1
Comparison of most frequent decision variables obtained in iterations *t*1 (100 simulation runs with uncertain EIBT) and *t*4 (one optimization run with known AIBT). Columns 2 and 3 displays the AIBT offset from the SIBT due to EIBT uncertainties and stand availability constraints. Columns 4 and 5 exhibit (the range of) departure delay. Columns 6 and 7 show whether an aircraft received an arrival prioritization, whereby percentage values in parenthesis display how frequent a value is part of an optimal solution in iteration *t*1. Columns 8 and 9 show if and how frequent a quick-turnaround was assigned to the respective aircraft. Columns 10 and 11 reveal the optimal stand allocation with the respective stand number and frequency. Columns 12 and 13 display how many quick transfers were advised for transfer groups originating from an aircraft, while columns 14 and 15 show how many transfers connections have been cancelled.

A/C	Δ AIBT-SIBT (min)		Δ TOBT-SOBT (min)		Arrival Prio		Quick Turn		Stand Change to		Quick Transfer (#)		Cancel Connex (#)	
	t1	t4	t1	t4	t1	t4	t1	t4	t1	t4	t1	t4	t1	t4
1	[44; 61]	55	[0; 8]	2	–$^{(2\%)}$	–	–$^{(0\%)}$	–	n/a (fixed on A6)		-	-	3	3
2	[17;25]	21	[7;19]	10	–$^{(0\%)}$	–	1$^{(89\%)}$	1	R1$^{(53\%)}$	R1	-	-	-	-
3	[0; 9]	5	[11;24]	17	–$^{(18\%)}$	–	–$^{(11\%)}$	–	n/a (fixed on B2)		-	-	-	-
4	[8;16]	13	[0; 3]	0	–$^{(0\%)}$	–	–$^{(0\%)}$	–	R3$^{(28\%)}$	R3	-	-	-	-
5	[38; 50]	45	[0; 0]	0	1$^{(56\%)}$	1	–$^{(0\%)}$	–	n/a (fixed on C2)		1	1	1	1
6	[-1; 21]	2	[0; 17]	14	–$^{(0\%)}$	–	–$^{(0\%)}$	–	fixed on A5		-	-	-	-
7	[9;25]	14	[0; 10]	0	–$^{(0\%)}$	–	–$^{(0\%)}$	–	C1$^{(87\%)}$	C1	-	-	-	-
8	[-7; 39]	25	[0; 3]	0	–$^{(0\%)}$	–	–$^{(0\%)}$	–	n/a (fixed on R1)		-	-	-	-
9	[11;24]	15	[0; 3]	0	–$^{(0\%)}$	–	–$^{(5\%)}$	–	R2$^{(80\%)}$	R2	-	-	-	-
10	[10;23]	14	[0; 2]	0	–$^{(0\%)}$	–	–$^{(0\%)}$	–	–$^{(95\%)}$	–	-	-	-	-
11	[-19; 12]	1	[0; 0]	0	–$^{(0\%)}$	–	–$^{(0\%)}$	–	A4$^{(47\%)}$	A4	-	-	-	-
12	[0; 12]	6	[1;17]	7	–$^{(0\%)}$	–	1$^{(78\%)}$	1	n/a (fixed on A3)		-	-	-	-
13	[29; 47]	42	[4;18]	13	–$^{(13\%)}$	–	–$^{(46\%)}$	1	R2$^{(83\%)}$	R2	-	-	-	-
14	[-6; 30]	18	[0; 8]	3	–$^{(0\%)}$	–	–$^{(0\%)}$	–	n/a (fixed on A1)		-	-	-	-
15	[10;29]	20	[0; 0]	0	–$^{(0\%)}$	–	–$^{(0\%)}$	–	n/a (fixed on C1)		-	-	-	-
16	[-3; 8]	2	[0; 10]	4	–$^{(0\%)}$	–	–$^{(0\%)}$	–	n/a (fixed on R3)		-	-	-	-
17	; [5;27]	17	[1;17]	15	–$^{(0\%)}$	–	–$^{(48\%)}$	–	R1$^{(83\%)}$	R1	-	-	-	-
18	[-7; 16]	3	[0; 15]	9	–$^{(0\%)}$	–	–$^{(0\%)}$	–	n/a (fixed on A3)		1	1	-	-
19	[20;35]	29	[0; 14]	8	–$^{(11\%)}$	–	1$^{(100\%)}$	1	R2$^{(100\%)}$	R2	1	1	2	2
20	[7;23]	15	; [2;17]	11	–$^{(0\%)}$	–	–$^{(0\%)}$	–	A2$^{(98\%)}$	A2	-	-	-	-
	$\mathbf{\Phi}$ = 18.1	$\mathbf{\Phi}$ = 17.9	$\mathbf{\Phi}$ = 5.4	$\mathbf{\Phi}$ = 5.6	$\mathbf{\Sigma}$ = 1	$\mathbf{\Sigma}$ = 1	$\mathbf{\Sigma}$ = 3	$\mathbf{\Sigma}$ = 4	$\mathbf{\Sigma}$ = 9	$\mathbf{\Sigma}$ = 9	$\mathbf{\Sigma}$ = 3	$\mathbf{\Sigma}$ = 3	$\mathbf{\Sigma}$ = 6	$\mathbf{\Sigma}$ = 6

iteration, from which we determine the most frequent values of decision variables and compare them against counterparts in the next iteration. Between iteration *t*1 (longest LAT with high EIBT uncertainties) and *t*4 (post-operational with constant EIBTs = known AIBT), we find that in total only one out of 215 decisions (0.4%) has been made differently (additional Quick-Turnaround for aircraft 13, see Table 1).

As the algorithm is allowed to use inherent system buffers with full flexibility, average in-block delay is roughly three minutes higher than average EIBT values. This anomaly arises from the fact that four aircraft (especially those which arrive earlier than scheduled) need to wait on apron up to 25 min for their stand to be vacant. Nevertheless, the system demonstrates good adaptive and restorative resilience capacities, given that average departure delay would have been 31 min if all processes were operated normally and initial stand allocations as well as passenger connections were maintained and can be reduced to 5.4 min, whereby only one aircraft is delayed by more than 15 min on the outbound side (see column 5 in Table 1).

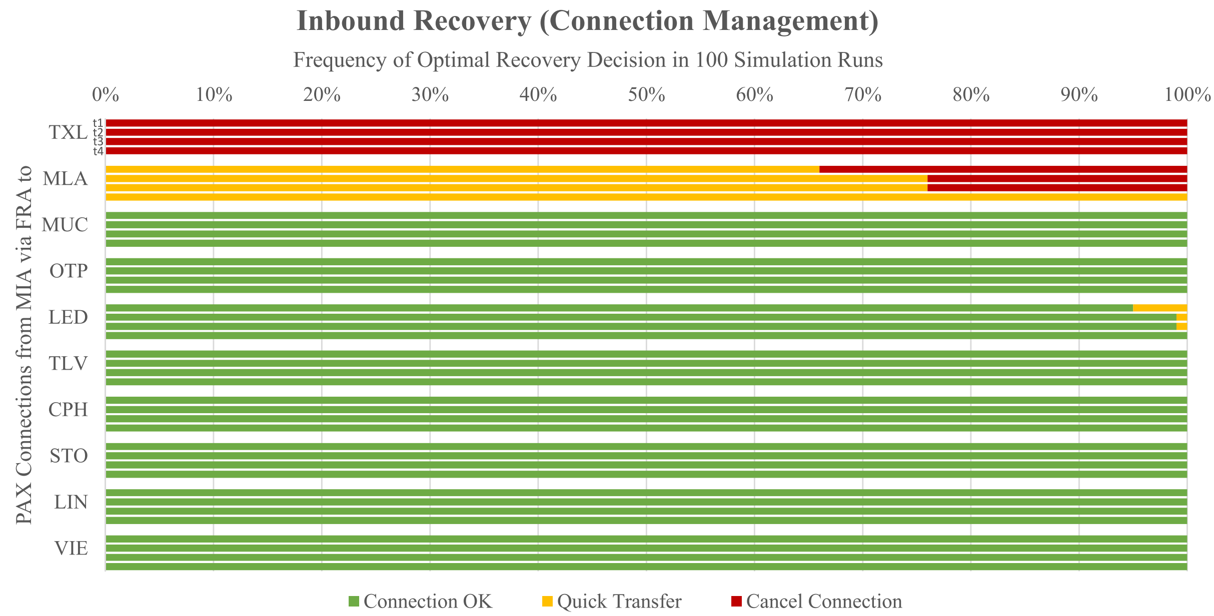
4.2. *Inbound recovery results*

Assessing potential inbound recovery options, we find that arrival prioritization (which is available only once for the entire period and reduces arrival delay by 5 min) is granted to aircraft 5 in 56% of all cases to reach the optimal solution. This makes sense when considering the great number of passenger connections coming from this long-haul arrival flight from Miami (MIA). As represented in Fig. 9 , one connection towards Berlin (TXL) is cancelled in all iterations t1-t4, given the 45-min inbound delay of aircraft 5. Another connection to Malta (MLA) is critical, but can be solved with the assignment of a quick passenger transfer in 63% of all cases in t1 (first bar). The alternative would be to cancel also this connection, so that a trade-off between departure delay cost and cost for rebooking passengers needs to made. With decreasing EIBT uncertainty in t2 and t3 (second and third bar), the "quick transfer" option has a higher frequency which corresponds to the solution from iteration *t*4 with known AIBT. All other connections are predicted to remain uncritical.

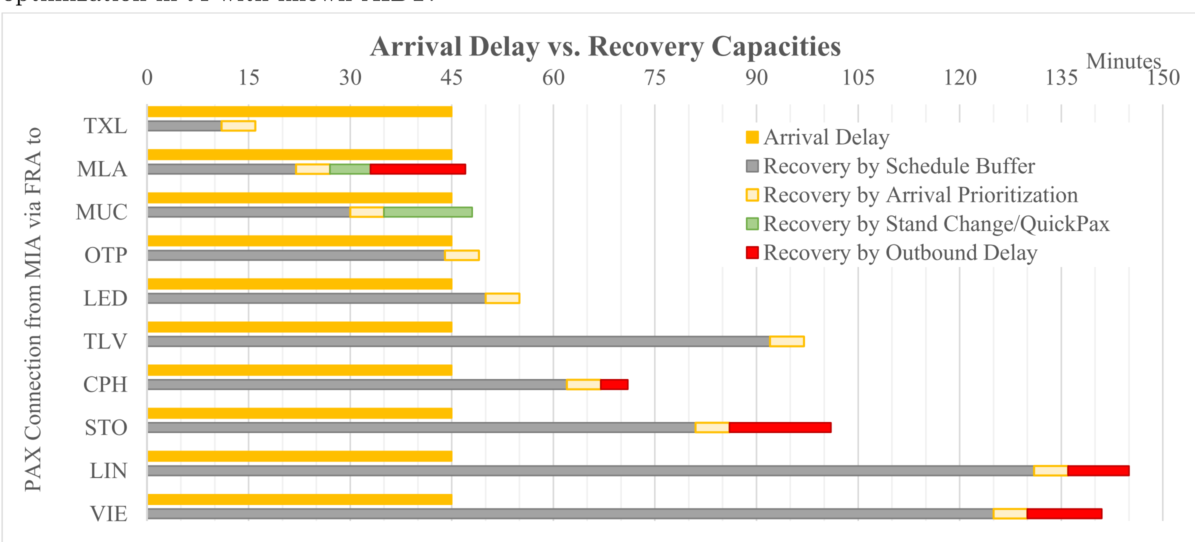
The numerical background regarding all transfer connections from this inbound flight is exhibited in Fig. 9b, so that the first connection to TXL would require additional 30 min of adaptive or restorative recovery in order to be maintained. The second connection to Malta is kept through the joint application of arrival prioritization, quick transfer and by delaying the departure flight. Note that the achieved recovery from all inbound recovery options is depicted as positive, while in reality their application decreases arrival delay and needed transfer times between aircraft. Fig. 9b and Table 1 further show that the re-allocation of aircraft 7 (going to MUC) to stand C1 (next to aircraft 5) saves the connection to MUC, while all other connections have enough buffer to absorb the initial arrival delay.

As mentioned above, only ten aircraft need to be considered for the stand re-allocation as all the others are fixed on their respective stands due to operational constraints. From these ten potential re-allocations, nine are made with a strong tendency towards positioning aircraft on remote positions. In fact and as detailed in Table 1, three aircraft are assigned to remote stand R2 and two aircraft are scheduled on remote stand R1 (in addition to aircraft 8 being fixed in stand R1). Both stands are preferred over remote stand R3 (one assigned aircraft), which is located further away from the bus station and thus requires longer transfer times (see Fig. 5). A benefit of remote stands are reduced deboarding and boarding durations, which directly impacts the total turnaround time and shows the close relationship between inbound and transit recovery.

As highlighted in Fig. 10, all re-allocations made in iteration *t*1 are confirmed in successive iterations *t*2 and *t*3 and within the post-operational optimization instance (*t*4). For most aircraft, the incidence for the most frequent assignment increases from *t*1 to *t*3 (see aircraft 7, 9, 13, 17, 20). Only for aircraft 10, the selected optimum in *t*1 had slightly less frequency in *t*2. Aircraft 2, 4 and 11, which arrive before 8:15 a.m. (*t*2), show the highest decision uncertainty which might be explained by a larger set of available stands in the early part of the scenario and higher EIBT uncertainties of all other flights. Nevertheless, the most frequent solutions selected in *t*1 equal the results from *t*4, although, as in the case of aircraft 4, the selected stand was only



(a) Frequency that selected recovery decision is optimal (first three bars contain results from 100 simulation runs with EIBT uncertainty in $t1$, $t2$ and $t3$, whereas the fourth bar displays the result from the single optimization in $t4$ with known AIBT.



(b) Comparison of arrival delay with system recovery capacities per passenger transfer connection. If arrival delay exceeds recovery capacities, the transfer is cancelled (see (a) above). Schedule buffers are given by the flight plan, while other recovery capacities are assigned by the schedule recovery model.

Fig. 9. Inbound recovery results exemplary for all passenger connections coming from the arrival flight with origin Miami (MIA).

assigned in 28% of all simulation runs.

4.3. *Transit recovery results*

As mentioned before, transit recovery aims at minimizing the ground time of an aircraft. The best solution to do so is to combine a quick-turnaround with an allocation to a remote stand, as was done for three out of four aircraft for which a quick-turnaround was assigned. As part of the quick-turnaround recovery option, a unit of additional cleaning, catering and loading agents is available to rotate between turnarounds in order to speed up operations. Quick-turnaround procedures are assigned to aircraft 2, 12 and 19 across all iterations. Also aircraft 13 and 17 showed high potential (46% and 48%) but are neglected in iteration $t1$ (see Table 1). The post-operational analysis

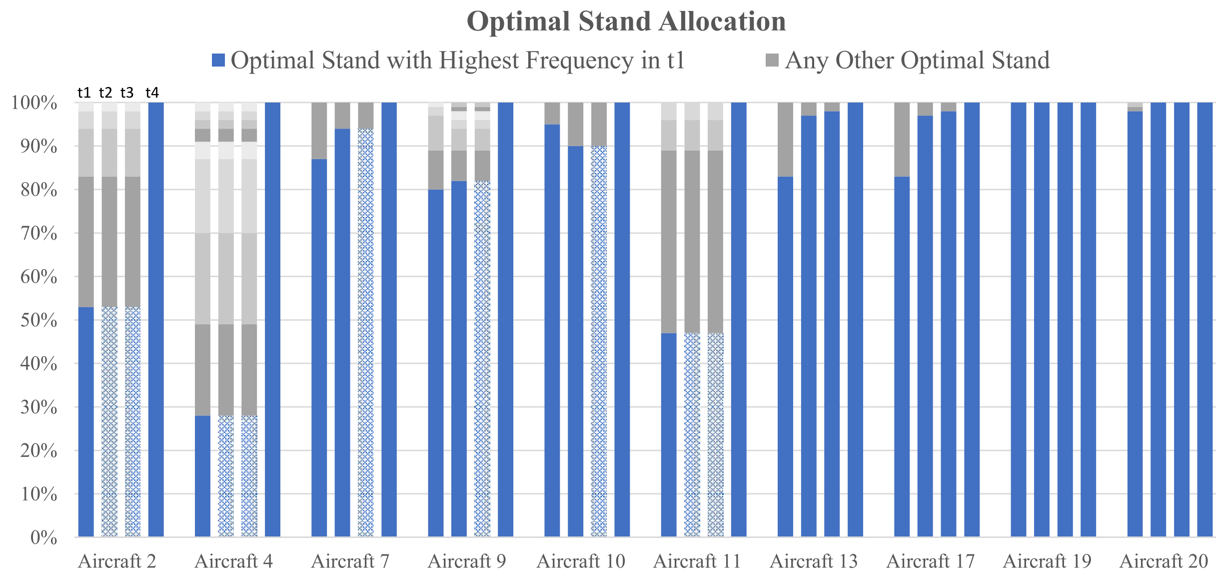


Fig. 10. Optimal stand allocation calculated in iterations $t1$, $t2$, $t3$ (100 simulation runs with EIBT uncertainty) and $t4$ (one optimization run with known AIBT) for those ten aircraft which have no operational constraints and can be assigned freely. Crossed bars represent fixed decisions for aircraft which have already arrived in this iteration).

shows that a quick-turnaround would have been optimal for aircraft 13 but not for 17, confirming somewhat that the decision was "50:50".

4.4. Outbound recovery results

Overall, only six out of 145 passenger connections (almost 4%) need to be cancelled while in most cases departure aircraft are slightly delayed, so that passengers can still reach their booked connections. Crew transfers are not affected in this case study, as the pair of aircraft which exchanged crews showed similar on-time performance.

5. Conclusion

The aim of this article was to analyse the impact of arrival time uncertainties on the stability of decision variables produced with an integrated schedule recovery model for airline ground operations. Within a case study setting of 20 aircraft arriving during a morning peak at FRA airport, an iterative optimization algorithm is applied to calculate the minimum recovery cost for a given set of arrival delays. Aircraft with longer look-ahead times to arrival are assumed with higher EIBT deviations, so that sensitivity of optimal recovery solutions is studied under uncertainty.

The analysis of the case study shows a high stability (thus, low sensitivity) of assigned schedule recovery solutions. Aggregated across all 100 simulation runs, only one recovery decision is obtained differently under the influence of uncertainty in comparison to an optimization which incorporates the full situational awareness of actual arrival times (AIBTs). This comes without any efforts of implementing robustness into the model further than the strategic schedule buffers which were already comprised in the adopted flight plan. Given the modelling approach with RCPSP, attention should be paid on bottleneck resources (such as stands in our case), such that jobs (aircraft) with lower priorities (low delay cost or high schedule buffers) are not assigned with extensive waiting times. Future developments of the model may therefore introduce slot constraints related to individual aircraft and resources. Furthermore it needs to be considered, that despite decisions being cost-optimal based on the current EIBT predictions, the individual solutions per simulation run deviate from an post-operational optimum (when AIBTs are known). Thus, prediction accuracy should be increased or the model should be transferred into a stochastic optimization model (for example by using chance constraints as in (Asadi et al., 2020)) to directly incorporate parameter uncertainties into the optimization process.

Judging from the results, one can easily recognize the complexity of relations and constraints to be considered in the decision-making process, which is still elaborated manually in airline operations control centers. It proves that it is hard to differentiate between pure inbound, transit and outbound recovery, as optimal solutions typically include several options from different categories for a given schedule deviation. Considering that AOCC agents are often reluctant to apply decision support systems in their daily operational routine, the model might be applied initially in shadow-mode to verify common "rules-of-thumb" and derive general dispatching rules for various network constellations. This might be further supported by deep-learning algorithms to detect typical data patterns in the optimal recovery solutions for different situations or cases which contain variable data accuracy.

Limitations in the current procedure regard the availability of real operational data, which led to the assumption of standardized normal distributed EIBTs. We are aware that timestamp predictions have individual distributions for each flight on different days and that the final timestamps (like AIBT) might be already subject to recovery decisions which we have no knowledge about. Likewise, passenger and crew itineraries are freely simulated so that they heavily influence the outcome of the scenario, as they do in everyday operations. Furthermore, we propose an airline-centric approach toward collaborative decision making, which includes the prerequisite that local stakeholders constantly share their resource availability and priorities with the airline (which is still not the case at many airports). We also neglect short-term disturbances related to some turnaround processes, like machine breakdowns or passenger no-shows, as the analysis of a second independent variable would exceed the scope of this paper. In any case, all mentioned limitations regard input parameters to our modelling approach, so that the itself is freely adaptable to the operational conditions and cost valuation of different airlines at various airports.

Future research may go in four different directions:

1. application of the algorithm to other days to assess if solution stability is generally guaranteed;

2. modelling of detailed passenger, crew, aircraft and maintenance schedules at downstream airport(s) in order to build flight-specific delay cost functions as proposed in (Evler et al., 2020);
3. incorporation of ATC restrictions like ATFM departure slots or arrival sequences in alignment to procedures such as enhanced slot-swapping or the user-driven prioritization process (Pilon et al., 2016);
4. further development of the integrated schedule recovery algorithm into a solution heuristic to be operational in real-time setting with variable user parameters (i.e., confidence level, available resources, etc.).

The recovery option of arrival prioritization was included in this paper to showcase that also recovery option applied during aircraft block times can be integrated into the model (as in (Delgado et al., 2016; Montlaur and Delgado, 2017), which would reduce the initial arrival delay). Thereby, we want to highlight that our focus on ground operations is complementary to other AOCC decision-making models being produced for tactical schedule and aircraft recovery.

Author statement

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